

Wilson Gichuhi

AI Systems & Reliability Engineer

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Professional Summary

AI systems engineer focused on making AI-driven products reliable in production. I architect the control layer between model outputs and real business workflows, replacing nondeterministic agent loops with explicit orchestration, auditable state, and evaluation infrastructure that catches failures before they reach users. I work best in high-stakes workflows where "mostly works" is still failure.

Professional Experience

AI Evaluation Engineer (Contractor), Bespoke Labs [🔗](#)

03/2026

- Contributed task authorship to OpenThoughts, Bespoke Labs' open reasoning dataset initiative that produced state-of-the-art small reasoning models trained on curated chain-of-thought data in collaboration with Stanford, Berkeley, and UT Austin.
- Designed and authored Terminal Bench style agentic coding tasks in the Harbor format, covering concurrency bugs, semaphore leaks, and write-ordering failures across real-world TypeScript and healthcare system scenarios, with full pipeline ownership from environment setup to oracle verification.
- Applied Bespoke-MiniCheck for claim-level factual grounding in evaluation pipelines, using NLI-based verification over LLM-as-judge alone to improve signal reliability in automated test suites.

Builder & AI Systems Engineer, Promco(Founder)

03/2026

- Building Promco, an order-to-approval system for liquor distributors that turns WhatsApp orders into commercially reviewable drafts before stock, margin, promo, and compliance decisions are committed.
- Designed the orchestration layer with explicit state machines and durable session handling, replacing free-form agent loops with traceable, recoverable workflow execution for compliance-sensitive order flows.
- Built guardrails around AI-assisted order processing using typed tool interfaces, schema validation, prompt/version control, and failure tracing so the system is debuggable and safe to run in production.
- Developed the product-resolution layer for noisy real-world order language(Kenyan slang), combining retrieval, ranking, and structured validation to improve SKU matching on colloquial WhatsApp orders.
- Built MCP-based integrations and runtime services over downstream commerce/compliance systems so models operate against live business context rather than ungrounded text generation.

AI Training & Model Evaluation Engineer, AfterQuery

08/2025

- Frontier Model Benchmarking: Conducted systematic evaluations of frontier LLMs for logic correctness and security; engineered automated test suites to identify vulnerabilities across domains.
- SWE-Bench Style Engineering (Project Anvil): Built end-to-end evaluation environments using base snapshots and gold-patch verification; automated oracle verification pipelines for complex domains including HL7 health-tech integrations and secure build systems.
- Terminal Bench v2: Built real-world agentic coding tasks targeting concurrency bugs, security vulnerabilities, and infra debugging — with full pipeline ownership from gold patches to oracle verification and pytest suites in containerized environments.
- Adversarial Engineering: Designed stress-tests for LLM-driven API design, moving beyond naive prompting to automated red-teaming of model-generated system architectures.

Backend Engineer, Alcora (FMCG Operations Platform)

02/2025 – 09/2025

- Architected Headless ERP as a multi-tenant backend with real-time stock and demand intelligence across distributed FMCG actors.
- Built core payment rails with automated debt reconciliation across brands, distributors, and retailers.
- Embedded geospatial intelligence for delivery tracking, route batching, and dispatch automation.
- Enabled promo cost vs incremental gross profit tracking (North Star Metric ≤ 0.6), reducing promotional wastage by 40%.

Founding Engineer, AfyaTelemed

05/2024 – 07/2025

- Architected a HIPAA-compliant telemedicine platform from scratch, covering triage, consultations, pharmacy, and lab integration.
- Built secure video consultation with E2E encryption (WebRTC) and payment processing across local and international gateways.
- Stack: Python/FastAPI, Next.js, WebRTC, AWS

Software Engineer, McSystems & Medical Inc.

04/2024 – 10/2024

- Architected a CRM with voice-to-CRM logging, reducing manual data entry by 70%.
- Rebuilt a legacy PHP medical sales marketplace into an offline-first Next.js platform.
- Designed a unified API layer integrating CRM and marketplace systems.

Fullstack Developer, JHUB Innovation Africa

10/2023 – 05/2024

- Built digital trade and climate-smart agriculture platforms within a Microsoft-partnered innovation ecosystem
- Delivered enterprise-grade authentication and workflow integrations across multi-stakeholder systems.

Technical Skills

Cognitive Architecture & Agentic Systems

- **State-Driven Logic:** XState (Finite State Machines), Apache Burr, Model Context Protocol (MCP).
- **Orchestration Patterns:** Agentic RAG, Multi-Agent Coordination, Tool-use, 12-Factor Agents.
- **Reliability Layer:** Effect-TS (Errors-as-Data), Circuit Breakers, Event-Driven Architectures.

Retrieval Diagnostics & Information Retrieval (IR)

- **Hybrid Search:** Vector + BM25, Rerankers, Atomic Fact Extraction
- **Data Stores:** Qdrant, Pinecone, pgvector (PostgreSQL), DuckDB.
- **Search Optimization:** Causal reasoning for retrieval drift, query expansion, and semantic caching.

Product & Frontend

- React & Tanstack Ecosystem
- PWAs, offline-first architecture
- Agent UI, HITL dashboard design

Evals & System Governance

- **Diagnostic Frameworks:** DeepEval, RAGAS, NLI (Natural Language Inference).
- **Safety & Alignment:** LLM-as-a-Judge, Automated Hallucination Detection, Evals-as-Code.
- **Synthetic Data & Robustness:** Curator (Synthetic Data Generation) for edge-case discovery, adversarial stress testing, and dataset augmentation.

Inference Infrastructure and Performance

- **Vitals & Optimization:** PagedAttention, Quantization (GGUF, AWQ, EXL2).
- **Resilient Compute:** Docker, Kubernetes (K8s), IaC, Cloudflare Workers (Edge Inference), Ollama.

Browser Automation & Web Agents

- Chrome DevTools Protocol (CDP), Playwright, Stagehand
- Agent action recording, session replay, DOM change detection
- Fault-tolerant navigation in auth flows and dynamic interfaces

Education

Bsc. Computer Technology, JKUAT University

2018 – 2023